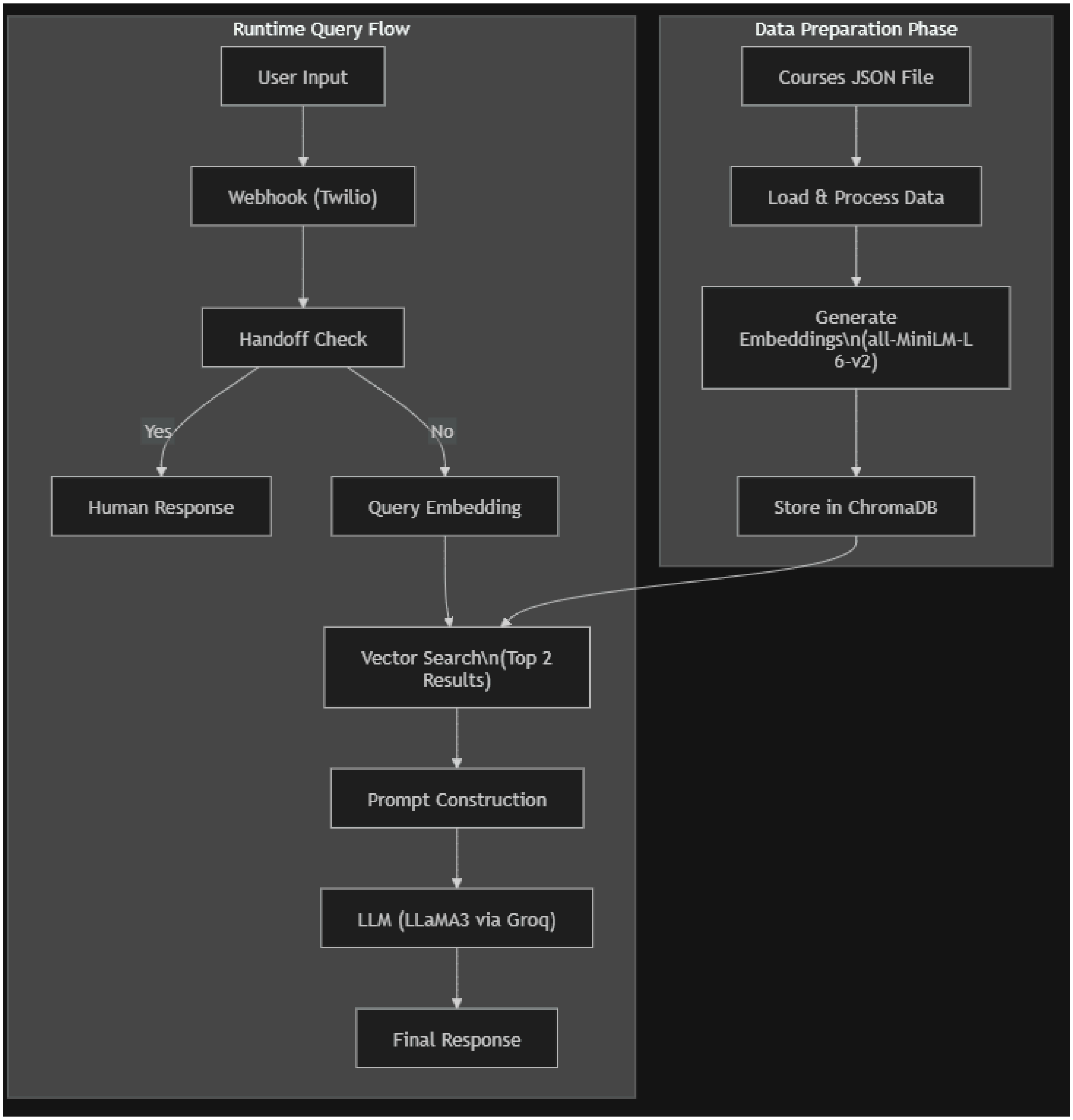
Architecture Diagram



Architecture Diagram Explanation

The architectural diagram above illustrates both the data preparation phase and the per-query retrieval-augmented generation flow:

1. Data Ingestion & Storage (Preparation Phase)

- **Data Source (`courses.json`):** Your detailed technical courses are stored as a structured JSON file.
- **Data Load & Process:** The `database_builder.py` script loads this JSON data. Each course's details are combined into a comprehensive text document.
- **Encoding (sentence-transformers):** The `all-MiniLM-L6-v2` model converts each course document into a mathematical vector embedding representing its semantic meaning.
- **Vector Database (ChromaDB):** These generated embeddings, along with the original text (documents) and course names (metadata), are stored in the locally persistent `chroma_data` directory.

2. Querying & Response Generation (Runtime Flow)

- **User Input:** The assistant receives a message (typically via the Twilio webhook in `main.py`).
- **Human Handoff Check (ai_engine.py):** Before any processing, the message is checked for specific handoff keywords (`'talk to human'`, `'real person'`, `'support'`).
    - **Handoff Branch:** If keywords are found, the system *immediately* bypasses the RAG logic and returns the pre-configured human counselor message. This flow terminates here for this query.
- **RAG Branch (No Handoff):** If no handoff is triggered, the full RAG process proceeds:
    - **Encoding (sentence-transformers):** The user's query is embedded using the *same* `all-MiniLM-L6-v2` model, creating a query vector.
    - **Vector Search & Retrieval:** This query vector is used to search the `courses_collection` in ChromaDB for the most semantically similar documents (`n_results=2`).
    - **Prompt Construction:** The retrieved relevant course information is combined with the user's original query into a structured prompt. This prompt includes explicit instructions (answer *only* from context, ask for email if interested, politely state if unknown).
    - **Large Language Model (Groq/Llama):** The constructed prompt is sent to the `llama3-70b-8192` model via the Groq SDK. The LLM generates a response based *strictly* on the provided context.
    - **Final Response:** The AI-generated response is sent back to the user (via the webhook and Twilio).

Block Diagram Showing Overview of entire application flow